# Day 1 Updates of team Bombardilo Crocodilo

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**On day 1, we interpreted the problem statement and noted down the requirements. We started with determining most suitable configuration of drone to be designed to tackle the mission. Then we did basic flight physics involving thrust, lift, and cruise speeds, accounting for changing conditions due to changing terrain and altitude as well. We also created a rough roadmap of the avionics suite to be fitted in the drone and made a rough weight distribution of the drone.**

## I. Problem Statement

We are assigned to make a logistics drone for a hilly region. It is supposed to replace porters and mules in steep terrain by carrying 10 kg to an altitude of 3,000 m. One way trip is 8km. We have to account for loss of thrust and endurance at an altitude also due to reduced density of air at higher elevations. There is existence of ridge turbulence of upto 12 m/s. We have to make our drone follow terrain for its flight path. Take off weight is under 25 kg .

## II. Configuration

A conventional quadcopter allows great accuracy for take off, landing and payload delivery. But such drones have poor range due to low efficiency. Fixed wing drones provide exceptional efficiency and allows great ranges but they require a proper infrastructure (flat surface for landing and takeoff) which is practically impossible at hilly terrains.
To combine best of both worlds, we will use hybrid Quadplane-VTOL configuration. It will takeoff vertically, then fixed wing rotor will start and when enough lift is generated by wing, the VTOL rotors will turn off to maximise efficiency. When drone is at target location, VTOL rotors again kicked in, drone starts hovering and delivery is done via winch mechanism.

# III. General Flight Physics

## A. VTOL Thrust

The following section outlines general flight physics.

Density of air at sea level

$$\rho_0 = 1.225\,\text{kg/m}^3$$

Density of air at 3000 m

$$\rho_{3000} = 0.909\,\text{kg/m}^3$$

$$\text{reduction} = \frac{\rho_0 - \rho_{3000}}{\rho_0} \times 100 = 25.8\%\ \text{drop}$$

$$\text{T} \propto \rho \quad \text{(VTOL thrust is directly proportional to air density)}$$

$$\text{therefore,} \quad \frac{T_0 - T_{3000}}{T_0} \times 100 = 26\% \quad \text{(For safer side we take 26\%)}$$

To counter turbulence, we specify the thrust-to-weight ratio at 3000 m ($\text{TW}_{3000}$):

$$\text{TW}_{3000} = 2:1 \quad \text{(after reduction in thrust)}$$

Hence, at sea level:

$$\text{TW}_0 = \frac{2}{1 - 0.258} = 2.7:1$$

$$\text{Maximum Takeoff Weight (MTOW)} = 25\,\text{kg}$$

$$\text{Therefore, thrust required} = 2.7 \times 25\,\text{kg}$$

$$= 67.5\,\text{kg}$$

### B. Fixed Wing Lift

$$L = \frac{1}{2}\rho V^2 S C_L$$

Lift must be same at sea level and 3000m:

$$\frac{1}{2}\rho_{3000} V_{3000}^2 S C_L = \frac{1}{2}\rho_0 V_0^2 S C_L$$

$$\implies V_{3000} = \sqrt{\frac{\rho_0}{\rho_{3000}}} V_0 = \sqrt{\frac{1.225}{0.909}} V_0$$

$$\implies V_{3000} = 1.16\, V_0$$

Theoretical % increase = 16%

$\therefore$ For safety, cruise airspeed is 17% higher at altitude.

## IV. Avionics and Sensor Suite Architecture

### A. Flight Controller

Pixhawk 6C

It has built in sensors like gyroscopes, accelerometers, barometers, and magnetometer and can easily handle computation like switching from VTOL mode to fixed wing mode mid flight

### B. Firmware

Ardupilot

It has built in libraries for UAV and is highly compatible with Pixhawk.

### C. Navigation

We use point GNSS module with RTK (real time kinematics).

Two antenna ensure no error due to intereference with magnets in motors and provide precise heading vector.

### D. LiDar Altimeter

Lightware SF11/C

We use this to enable our drone to follow terrain as flight path. Range of 40-50m gives drone enough time to react to sudden terrain changes.

### E. Payload Release Mechanism

- DC Planetary geared motor with integrated optical sensor to provide high torque to lift 10 kg without stalling.
- Braided dyneema cable which has high tensile strength and feature practically zero stretch.
- Machined aluminium pulley
- S beam load cell sensor to measure tension in cable. The moment payload is kept at ground, tension falls to 0.
- Eltronically controlled mechanical servo release latch releases the payload the moment measure tension falls to 0 and then cable is pulled up to winch motor

All of these electronics can be connected to CAN bus port of flight controller or companion microcontroller like ESP 32.

### F. Telemetry

Long range telemetry at 433 MHz is used for real time monitoring of battery voltage, flight mode, GPS lock, payload status etc.

## V. Weight Distribution

We made rough calculation of weight distribution

- Payload = 10 kg
- Battery pack = 3 kg
- Airframe and structures (mostly carbon fiber/composite) = 6 kg
- Rotors = 3 kg
- Avionics suite = 1.5 kg